Lecture 5: Dirichlet's Theorem and applications of the FTA

§ Theorem 1.21 (Dirichlet's theorem on primes in arithmetic progressions) Let $a, b \in \mathbb{Z}$ with $b > 0$ and $\gcd(a, b) = 1$. Then the arithmetic progression

$$a,\ a+b,\ a+2b,\ a+3b,\ \dots$$

contains infinitely many prime numbers.

§ Remark (on Theorem Theorem 1.21 (Dirichlet's theorem on primes in arithmetic progressions)) Setting $a = 1$ and $b = 1$ recovers the elementary statement that there are infinitely many primes. (We remark that the full proof of Dirichlet's theorem requires analytic tools and is beyond the scope of this course.)

§ Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4) There are infinitely many primes of the form $4n + 3$, where n is a nonnegative integer.

§ Lemma 1.23 (Product of numbers congruent to 1 modulo 4) Let $a, b \in \mathbb{Z}$. If $a \equiv 1 \pmod 4$ and $b \equiv 1 \pmod 4$, then $ab \equiv 1 \pmod 4$.

Proof. Write $a = 4n_1 + 1$ and $b = 4n_2 + 1$ with $n_1, n_2 \in \mathbb{Z}$. Then

$$ab = (4n_1 + 1)(4n_2 + 1) = 16n_1n_2 + 4n_1 + 4n_2 + 1 = 4(4n_1n_2 + n_1 + n_2) + 1,$$

so $ab \equiv 1 \pmod 4$. This proves the lemma. □

Proof of Proposition Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4). Suppose, for contradiction, that there are only finitely many primes congruent to 3 modulo 4. List them as

$$P_1 = 3,\ P_2,\ P_3,\ \dots,\ P_r,$$

all distinct and each $P_i \equiv 3 \pmod 4$.

Consider the integer

$$N := 4P_1P_2 \cdots P_r - 1.$$

Note that $N \equiv -1 \equiv 3 \pmod 4$.

Let q be any prime divisor of N. We make two observations:

1. For each $i = 1, \dots, r$ we have $N \equiv -1 \pmod{P_i}$, so $P_i \nmid N$. Thus q is not equal to any of the P_i.
2. Since $N \equiv 3 \pmod 4$, it is not possible that every prime divisor of N is congruent to 1 modulo 4. Indeed, a product of integers each congruent to $1 \pmod 4$ is itself congruent to $1 \pmod 4$ by Lemma Lemma 1.23 (Product of numbers congruent to 1 modulo 4), contradicting $N \equiv 3 \pmod 4$. Hence some prime divisor q of N satisfies $q \equiv 3 \pmod 4$.

Combining the two observations, q is a prime congruent to 3 modulo 4 that is not among $P_1, \dots, P_r$, contradicting the assumed finiteness of such primes. Therefore there are infinitely many primes congruent to 3 modulo 4. □

§ Remark (on Proposition Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4)) The method above—building an integer with residue $3 \pmod 4$ whose prime divisors cannot all be from a given finite list—gives a proof in this special case of Dirichlet's theorem. However, this argument does not extend directly to all arithmetic progressions; for the full theorem one needs deeper tools (Dirichlet characters and L-series).

Congruences

§ Historical remark (Gauss and congruences) Up to the beginning of the nineteenth century number theory consisted of many isolated results. In 1801, Gauss, in his *Disquisitiones Arithmeticae*, introduced the modern theory of congruences and unified many isolated results in number theory.

§ Definition (Congruence modulo m) Let $a, b, m \in \mathbb{Z}$ with $m > 0$. We say that *a is congruent to b modulo m*, and write

$$a \equiv b \pmod{m},$$

if $m \mid (a - b)$. The integer m is called the *modulus* of the congruence.

§ Remark (notation) The notation $a \not\equiv b \pmod{m}$ means $m \nmid (a - b)$. The modulus is customarily taken to be positive; we will assume $m > 0$ in the sequel.

§ Proposition 2.1 (Congruence modulo m is an equivalence relation) Congruence modulo m is reflexive, symmetric, and transitive on $\mathbb{Z}$; hence it is an equivalence relation.

Proof. Reflexive: For any $a \in \mathbb{Z}$, $a - a = 0$ and $m \mid 0$, so $a \equiv a \pmod{m}$.

Symmetric: If $a \equiv b \pmod{m}$ then $m \mid (a - b)$, hence $m \mid -(a - b) = (b - a)$, so $b \equiv a \pmod{m}$.

Transitive: If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$ then $m \mid (a - b)$ and $m \mid (b - c)$, therefore $m \mid (a - b) + (b - c) = a - c$, so $a \equiv c \pmod{m}$. □

§ Consequence 2.2 (Partition into residue classes) The set $\mathbb{Z}$ is partitioned into equivalence classes under congruence modulo m.

§ Definition (Notation for residue class) For $x \in \mathbb{Z}$, write $[x]_m$ (or simply $[x]$ when m is understood) for the equivalence class of x modulo m:

$$[x]_m = \{y \in \mathbb{Z} : y \equiv x \pmod{m}\}.$$

(Do not confuse $[x]_m$ with the floor function.)

Example: The equivalence classes modulo 4

$$\begin{aligned}
[0]_4 &= \{x \in \mathbb{Z} : x \equiv 0 \pmod{4}\} = \{4n : n \in \mathbb{Z}\} = \{\dots, -8, -4, 0, 4, 8, \dots\},\\
[1]_4 &= \{x : x = 4n + 1,\ n \in \mathbb{Z}\} = \{\dots, -7, -3, 1, 5, 9, \dots\},\\
[2]_4 &= \{x : x = 4n + 2,\ n \in \mathbb{Z}\} = \{\dots, -6, -2, 2, 6, 10, \dots\},\\
[3]_4 &= \{x : x = 4n + 3,\ n \in \mathbb{Z}\} = \{\dots, -5, -1, 3, 7, 11, \dots\}.
\end{aligned}$$

Thus $\mathbb{Z}$ is partitioned into four residue classes modulo 4, and every integer is congruent to exactly one of $0, 1, 2, 3$ modulo 4.

§ Definition (Complete residue system modulo m) A set $R \subset \mathbb{Z}$ is a *complete residue system modulo m* if every integer is congruent modulo m to exactly one element of R.

§ Example (complete residue systems) The set $\{0, 1, 2, \dots, m - 1\}$ is a complete residue system modulo m. Also, for example, $\{6, -11, 19, 1988\}$ is a complete residue system modulo 4, since $6 \equiv 2$, $-11 \equiv 1$, $19 \equiv 3$, and $1988 \equiv 0 \pmod{4}$.

§ Proposition 2.3 (Standard complete residue system) The set $\{0, 1, 2, \dots, m - 1\}$ is a complete residue system modulo m.

Proof. Let $a \in \mathbb{Z}$. By the Division Algorithm there exist unique $q, r \in \mathbb{Z}$ with $0 \le r < m$ such that $a = mq + r$. Then $a - r = mq$, so $m \mid (a - r)$ and $a \equiv r \pmod{m}$, with $r \in \{0, 1, \dots, m - 1\}$. This shows every integer is congruent to at least one element of $\{0, 1, \dots, m - 1\}$.

To show uniqueness: suppose $r_1, r_2 \in \{0, 1, \ldots, m-1\}$ and $r_1 \equiv r_2 \pmod{m}$. Then $m \mid (r_1 - r_2)$, but $-(m-1) \leq r_1 - r_2 \leq m-1$. The only multiple of m in this range is 0, so $r_1 - r_2 = 0$ and hence $r_1 = r_2$. Thus each integer is congruent to exactly one element of $\{0, 1, \ldots, m-1\}$. ☐

§ Proposition 2.4 (Arithmetic with congruences) Let $a, b, c, d \in \mathbb{Z}$ and $m > 0$. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

1. $a + c \equiv b + d \pmod{m}$,
2. $ac \equiv bd \pmod{m}$.

Proof. From $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ we have $m \mid (a-b)$ and $m \mid (c-d)$.

For (1): $m \mid (a-b) + (c-d) = (a+c) - (b+d)$, hence $a + c \equiv b + d \pmod{m}$.

For (2): Note $m \mid (a-b)$ implies $m \mid c(a-b)$, and $m \mid (c-d)$ implies $m \mid b(c-d)$. Add these divisibilities to get $m \mid c(a-b) + b(c-d) = ac - bd$, so $ac \equiv bd \pmod{m}$. ☐